

Stack Overflow Survey Report

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Report Outlines

Outlines:

- **Title**
- **Outline**
- **Executive Summary**
- **Introduction**
- **Methodology**
- **Programming Languages Trends**
- **Databases Trends**
- **Dashboards**
- **Insights from Dashboards**
- **Overall Findings and Implications**
- **Conclusion**



Executive Summary

Key Findings Summary Template:

- **Main Objective:** Project aimed at building a Dashboard to visualize current and future trends for Programming Languages, Databases etc. in Software Industry.
- **Key Insights:** HTML/CSS, Javascript, Python, C# etc are currently trending and same is expected for future. For Databases, SQL Databases are trending and same goes for the future.
- **Data Trends:** Programming languages like C#, Python, JS are rising at a great pace. Similarly, NoSQL Databases are also rising at fast pace. Platforms like AWS, Azure are in high-demand.
- **Challenges:** Aggregating data from multiple resources and ensuring accuracy and consistency of dataset was time-consuming.
- **Conclusions:** The Dashboard offers valuable insights into trends in Software Industry and also tells us about future trends as well.



Introduction

Report Purpose, Target Audience, and Value:

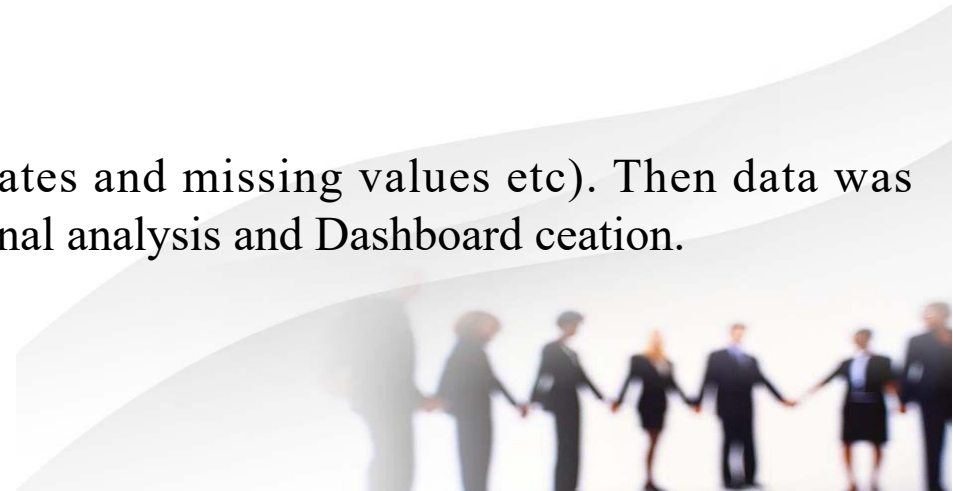
- **Purpose:**
 - The purpose of this report is to summarize the whole project and provide key findings and insights to the relevant stakeholders.
- **Target Audience:**
 - This report target wide range of audience including Software Engineers, Technology managers, Product Owners, Tech Startups, IT Consultants etc.
- **Value:**
 - This report aimed at aiding the target audience in informed decision-making, optimizing strategies, identifying trends and improving efficiency etc.



Methodology

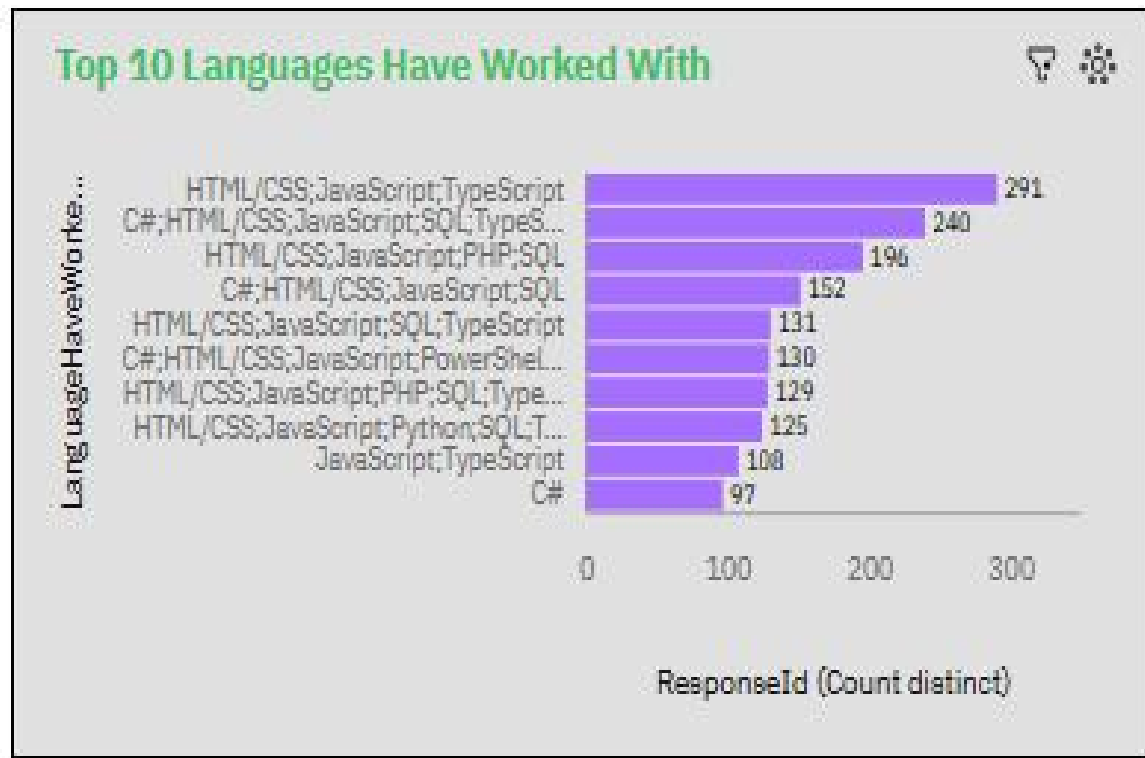
Data Sources, Collection Methods, and Key Wrangling Steps:

- **Data Sources:**
 - This dataset has come from Stack Overflow and it is a survey data conducted by Stack Overflow.
- **Collection Methods:**
 - This dataset is a standard CSV (Comma - Separated Values) files downloaded directly from Stack Overflow database.
- **Key Wrangling Steps:**
 - Data was properly cleaned and preprocessed (handling duplicates and missing values etc). Then data was normalized, standardized, transformed and finally aggregated for final analysis and Dashboard ceation.

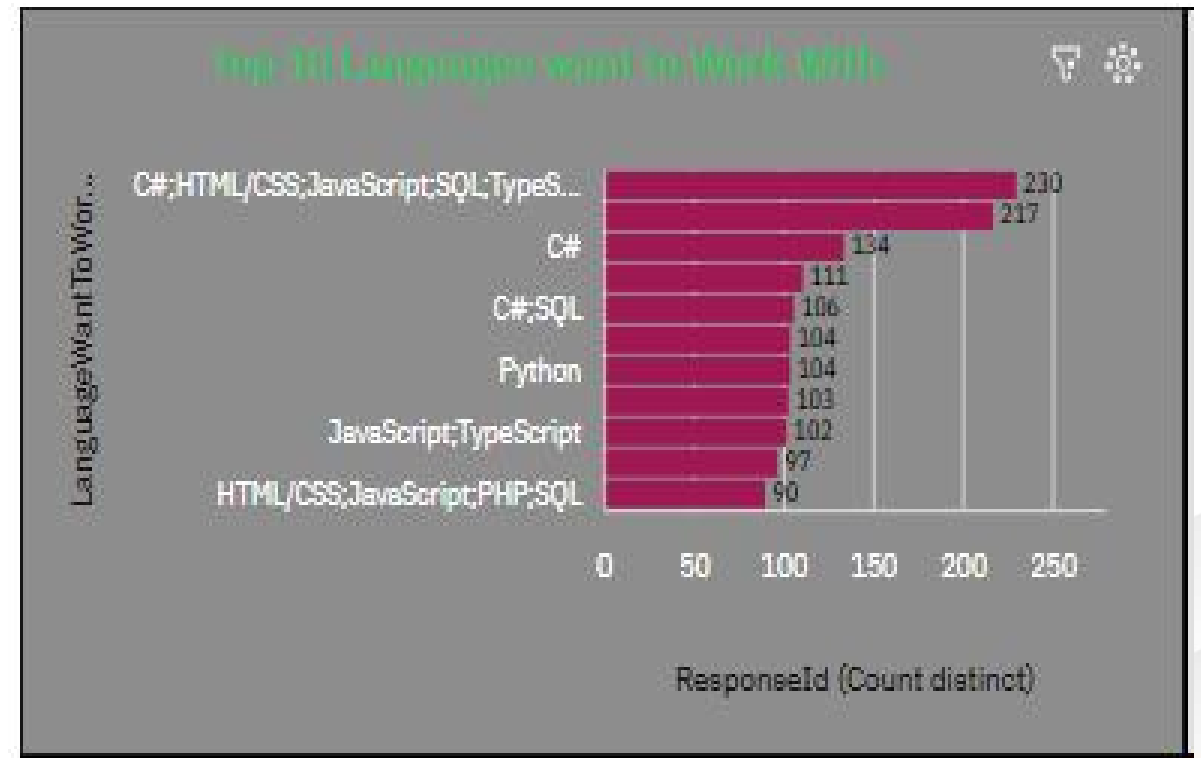


Programming Languages Trend - I

Current Year



Next Year



Programming Languages Trend - II

Findings:

- Languages like HTML, CSS, JavaScript, TypeScript, Python, C# etc are in high demand.
- Bar Chart for future trend shows that same programming languages are going to rule the trend in the next year or upcoming future as well.
- High demand of languages like HTML / CSS, JavaScript, TypeScript and Python shows that Web Development will be in rise in near future.

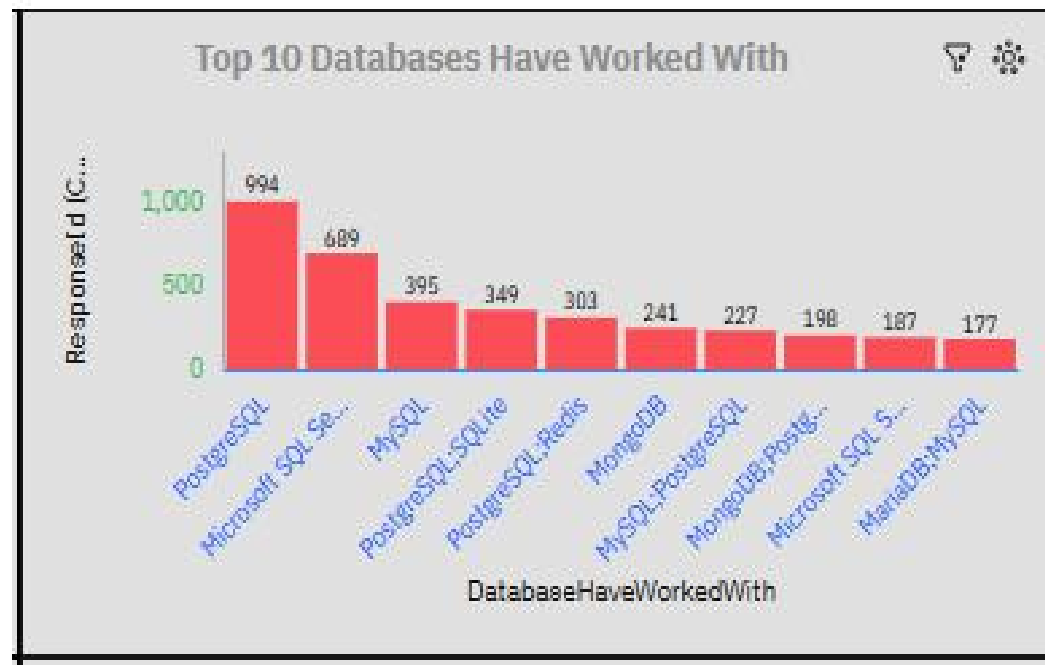
Implications:

- High demand of these languages suggest that Software Industry is more focused on Web Development.
- Future trends suggest that companies should act and recruit skillful peoples in relevent expertise to keep themselves in competition.
- Companies should be aware of upcoming future technologies and emerging skills.

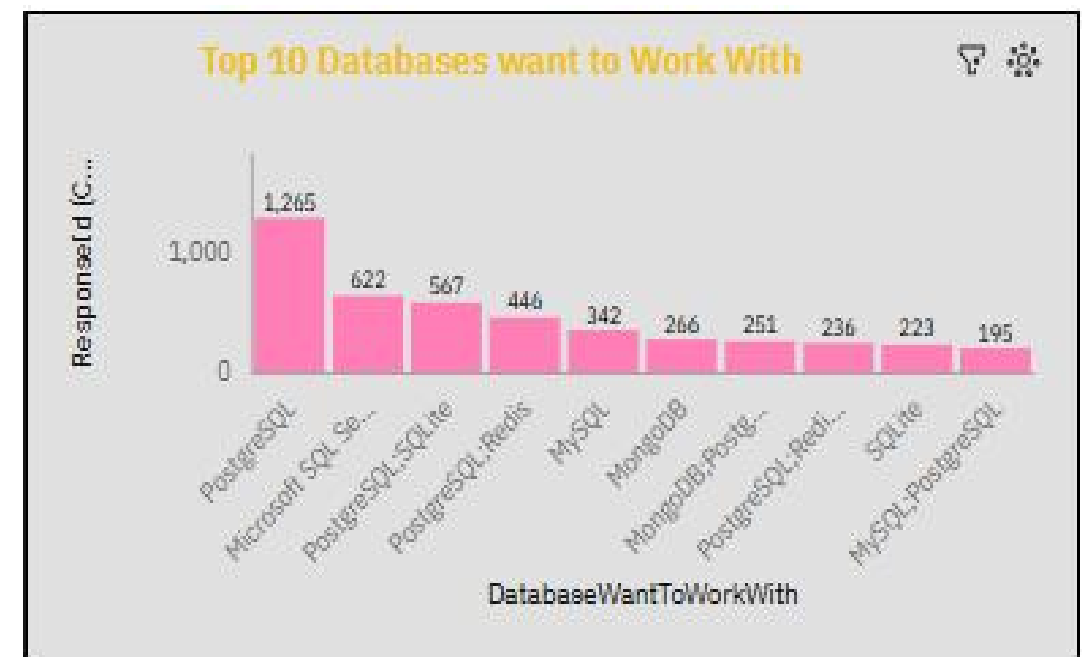


Database Trends - I

Current Year



Next Year



Database Trends - II

Findings:

- SQL Databases like PostgreSQL, Microsoft SQL Server, MySQL etc are in high-demand.
- Future trend shows that same SQL Databases will be trending and remain in demand in future.
- NoSQL Databases i.e MongoDB etc are also rising due to need of handling databases without SQL.

Implications:

- Developers need to lookout for this important skill so they must not left behind as SQL demand is rising,
- Future trends suggest that companies should act and recruit skillful peoples in relevent expertise to keep themselves in competition.
- Companies should not neglect the importance of NoSQL databases and must be aware of their trend.



Dashboard

Now I will present my Dashboard in the following slides.



Dashboard: Current Technology Usage

Current Technology Usage.

Future Technology Trend.

Demographics.

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Top 10 Languages Have Worked With



Top 10 Databases Have Worked With



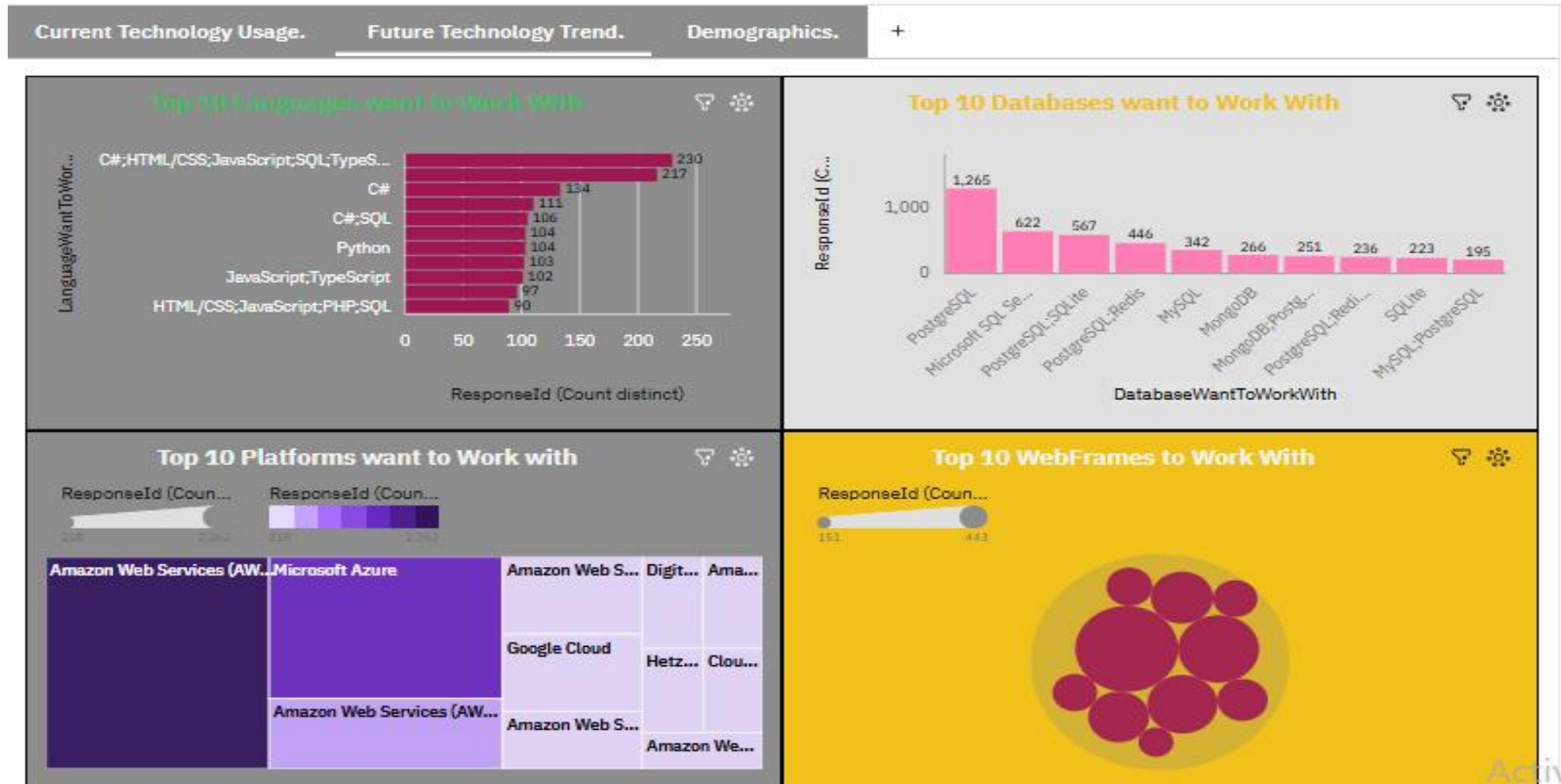
Top 10 Platforms have Worked With



Top 10 Frameworks have Worked with



Dashboard: Future Technology Trends



Dashboard: Demographics

Current Technology Usage.

Future Technology Trend.

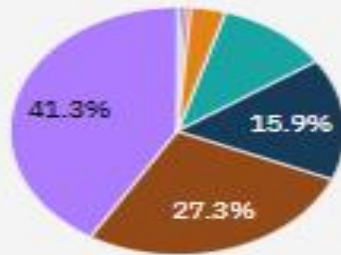
Demographics.

+

Respondent distribution by Age.

Age

- Prefer not to say
- 55-64 years old
- 35-44 years old
- 65 years or older
- 45-54 years old
- 25-34 years old
- Under 18 years old
- 18-24 years old



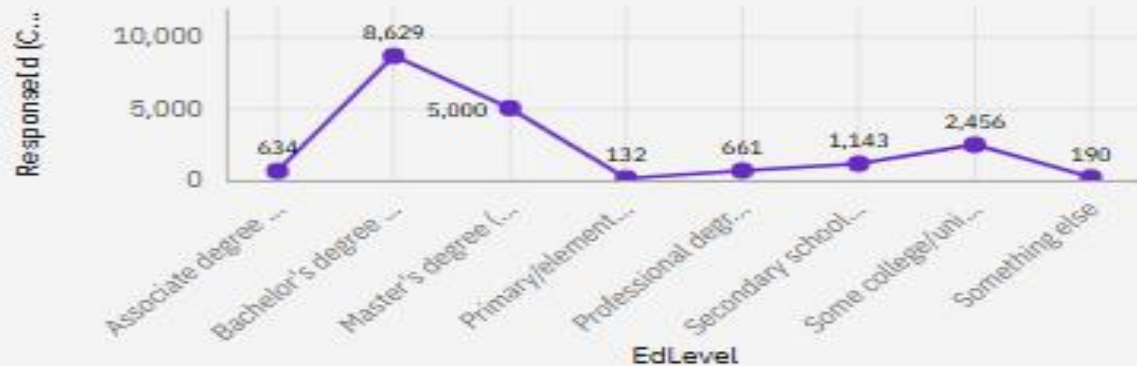
Respondent Count by Country.

ResponseId

3	10	11	12	19	23	24	29
33	75	113	122	140	150	166	173
181	183	190	223	235	299	315	320



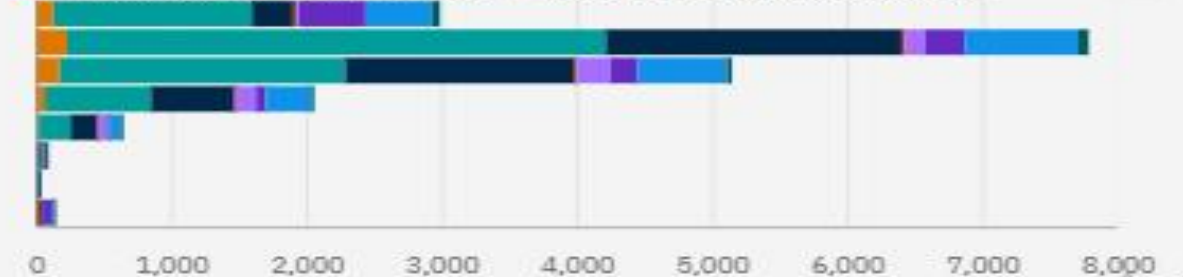
Respondent distribution by Formal Education Level



Respondent Count by Age, classified by Education Level

EdLevel

- Associate degree (A.A., A.S., etc.)
- Master's degree (M.A., M.S., M.En., etc.)
- Professional degree (J.D., M.D., Ph.D., etc.)
- Bachelor's degree (B.A., B.S., B.E., etc.)
- Primary/elementary school
- Secondary school (e.g., American high school)



Insights from Dashboard

Here's a concise summary of insights from Dashboard:

- **Programming Languages:**

- Languages like HTML / CSS, JavaScript, Python, SQL are currently popular. TypeScript and C# are also showing an increase in demand due to their usage in Web Development and Game Development etc. Similarly, these languages will remain in high-demand in future.

- **Databases:**

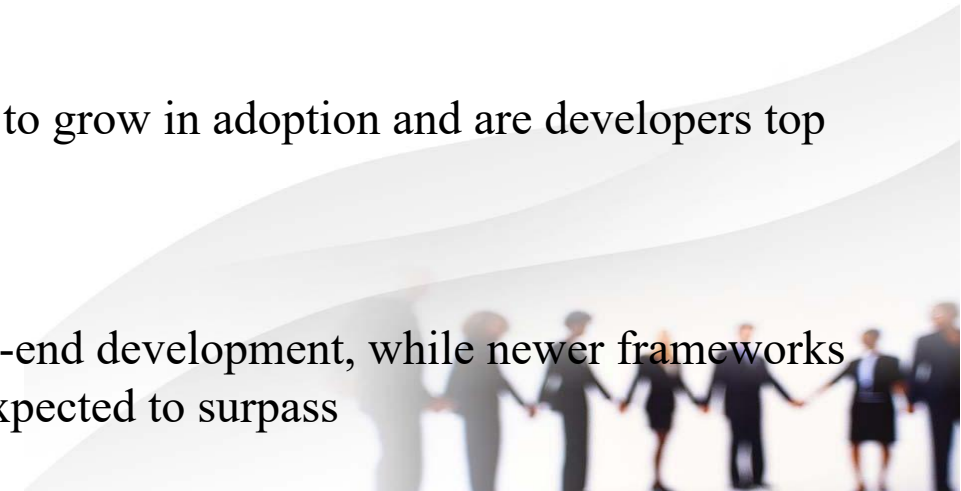
- SQL-based databases are currently dominating the trend, but NoSQL solutions like MongoDB are silently rising, for their unstructured data needs.

- **Platforms:**

- Cloud platforms, particularly AWS, Azure, and Google Cloud, continue to grow in adoption and are developers top choices.

- **Web Frameworks:**

- Spring Boot, React, ASP .NET Core hold steady as top choices for front-end development, while newer frameworks like FastAPI, Express.js, Node.js are gradually gaining attention, thus expected to surpass the latters in future.



OVERALL FINDINGS & IMPLICATIONS

Findings:

- Because of their adaptability in web development, machine learning, and game development, languages like Python, JavaScript, and C# continue to be the most sought-after.
- As the necessity for managing unstructured data grows, NoSQL databases like MongoDB are becoming more and more popular alongside conventional SQL databases.
- The foundation of modern IT infrastructure is progressively being formed by cloud platforms (AWS, Azure, and Google Cloud).

Implications:

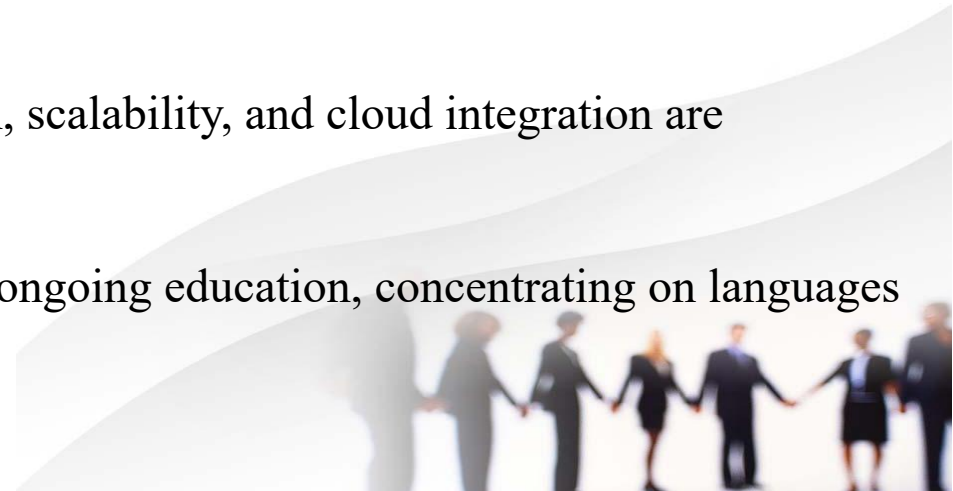
- To improve their employability in a variety of sectors, developers should concentrate on learning Python, JavaScript, and C#. Organizations can use these languages to create flexible teams.
- To support a variety of data types and increase data management flexibility, businesses should implement hybrid database systems.
- Cloud integration should be a top priority for businesses, and developers should acquire cloud-related skills to keep up with the industry's shift to distributed systems.



Conclusion

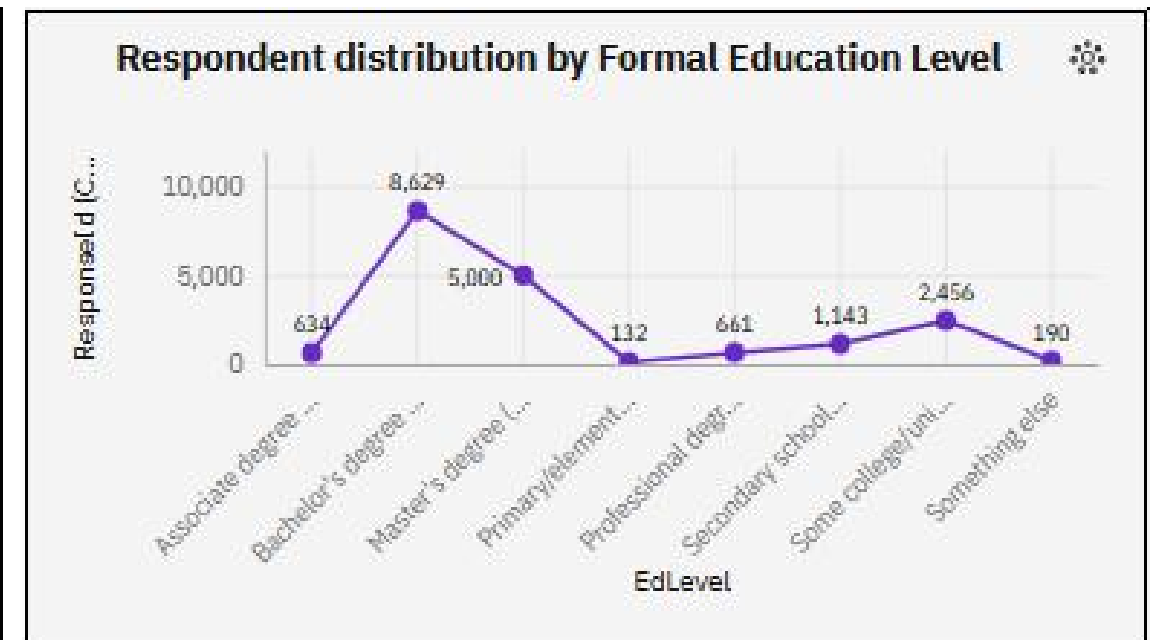
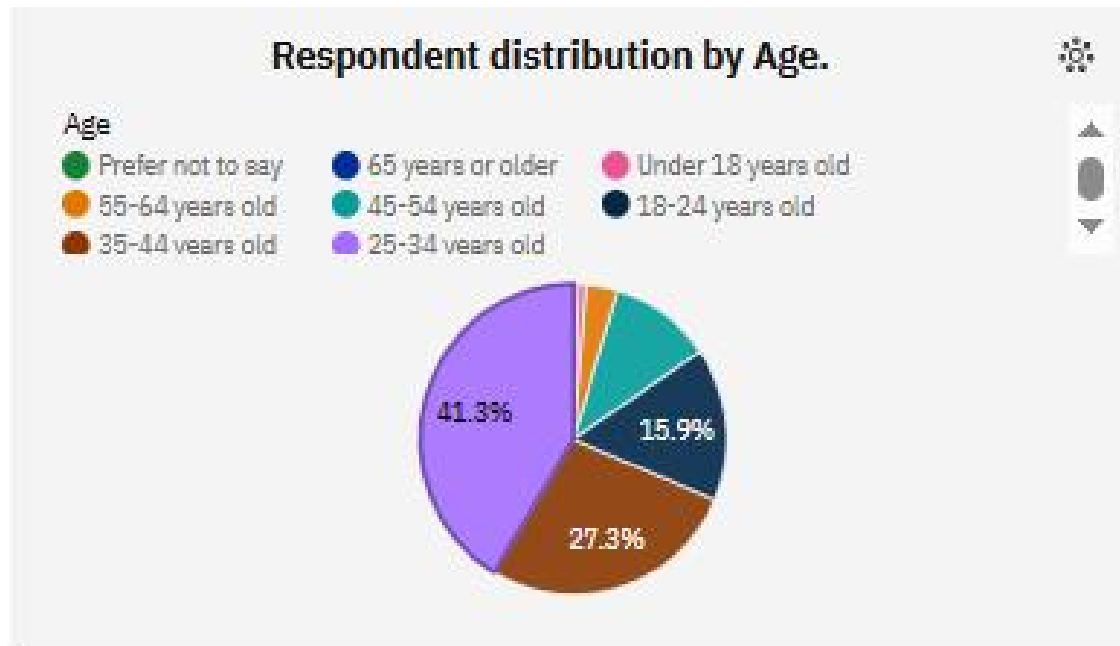
Project Findings:

1. **Dominant Languages:** Python, JavaScript, and C# continue to be in great demand because of their adaptability to machine learning, web development, and game development.
2. **Emerging Trends:** Cloud platforms and contemporary web frameworks are influencing future development methodologies, while TypeScript is becoming more popular for improving JavaScript apps.
3. **Database Preferences:** NoSQL solutions are increasingly being used to handle unstructured data, while SQL-based databases still hold a dominant position.
4. **Future Prospects:** Flexible languages and tools that facilitate automation, scalability, and cloud integration are preferred by the tech industry.
5. **Skill Development:** To keep ahead of the curve, developers must pursue ongoing education, concentrating on languages and technologies that reflect current market trends.



Appendix

Here is a Pie-chart and a Line-Chart representing the Age Distribution and Education Level Distribution of Respondents respectively.



JOB POSTINGS

In Module 1 you have collected the job posting data using Job API in a file named “job-postings.xlsx”. Present that data using a bar chart here. Order the bar chart in the descending order of the number of job postings.

